

— FINAL PROJECT • DECISION RECORD

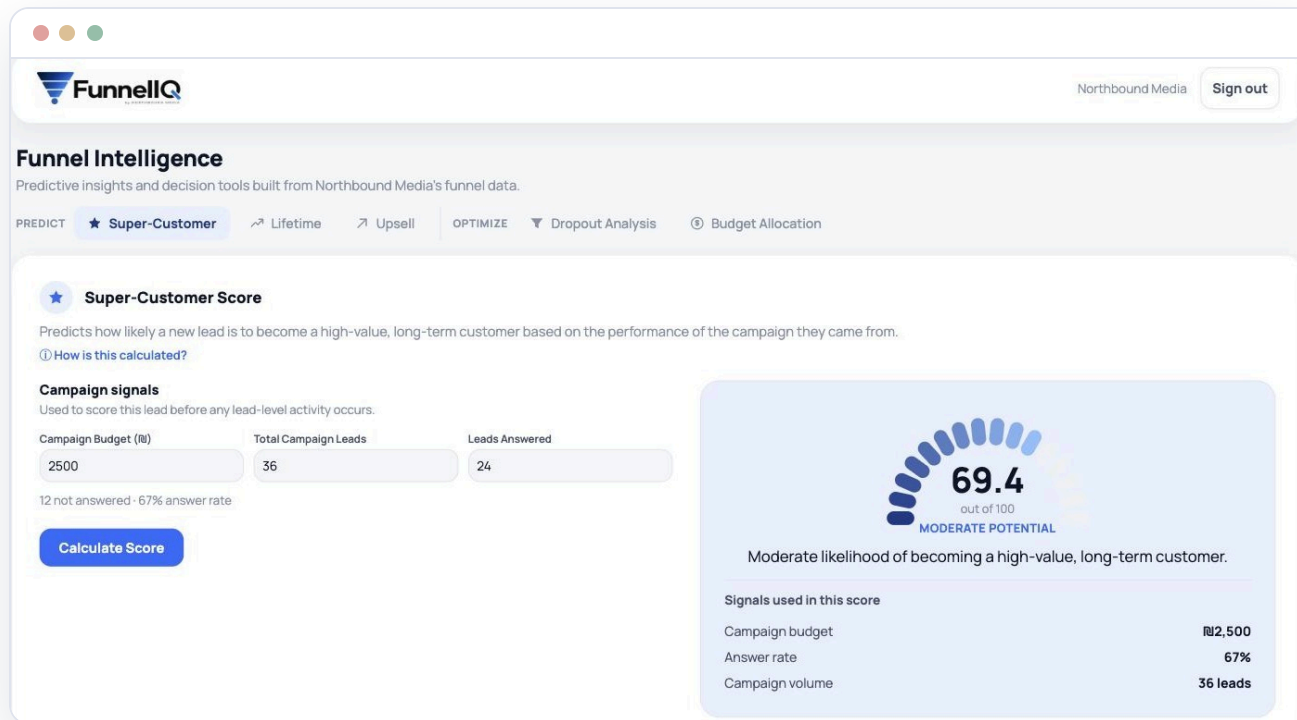
FunnelIQ

Turning 3,500 rows of historical funnel data into a deployed decision-support tool.



A deployed workspace for prediction and optimization

Five decision tools built on Northbound Media's campaign data.



— PREDICT

Super-Customer · Lifetime · Upsell

— OPTIMIZE

Follow-up Dropout · Budget Allocation

5 DECISION TOOLS

AUTHENTICATED

DEPLOYED

One dataset, five business questions

3,500 raw records → validation & cleaning → 3,490 analysis-ready records

PREDICT

- ★ How long will a customer stay?
- ↗ Who is likely to upsell?
- ↗ Which campaigns produce high-value customers?

OPTIMIZE

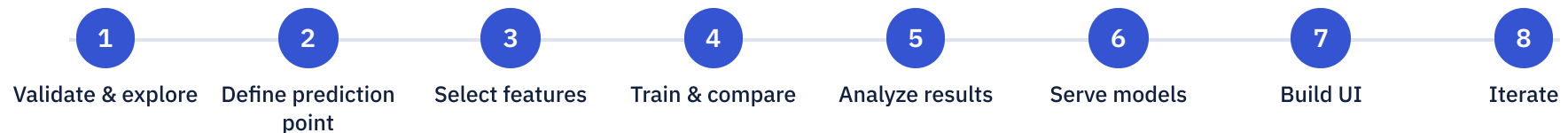
- ▽ How far should sales keep following up?
- ◎ How should ad budget be allocated?

EARLY SIGNAL

Mid-budget campaigns (€2,000–5,000) unexpectedly converted better than higher-spend campaigns. The same pattern re-emerges across other analyses.

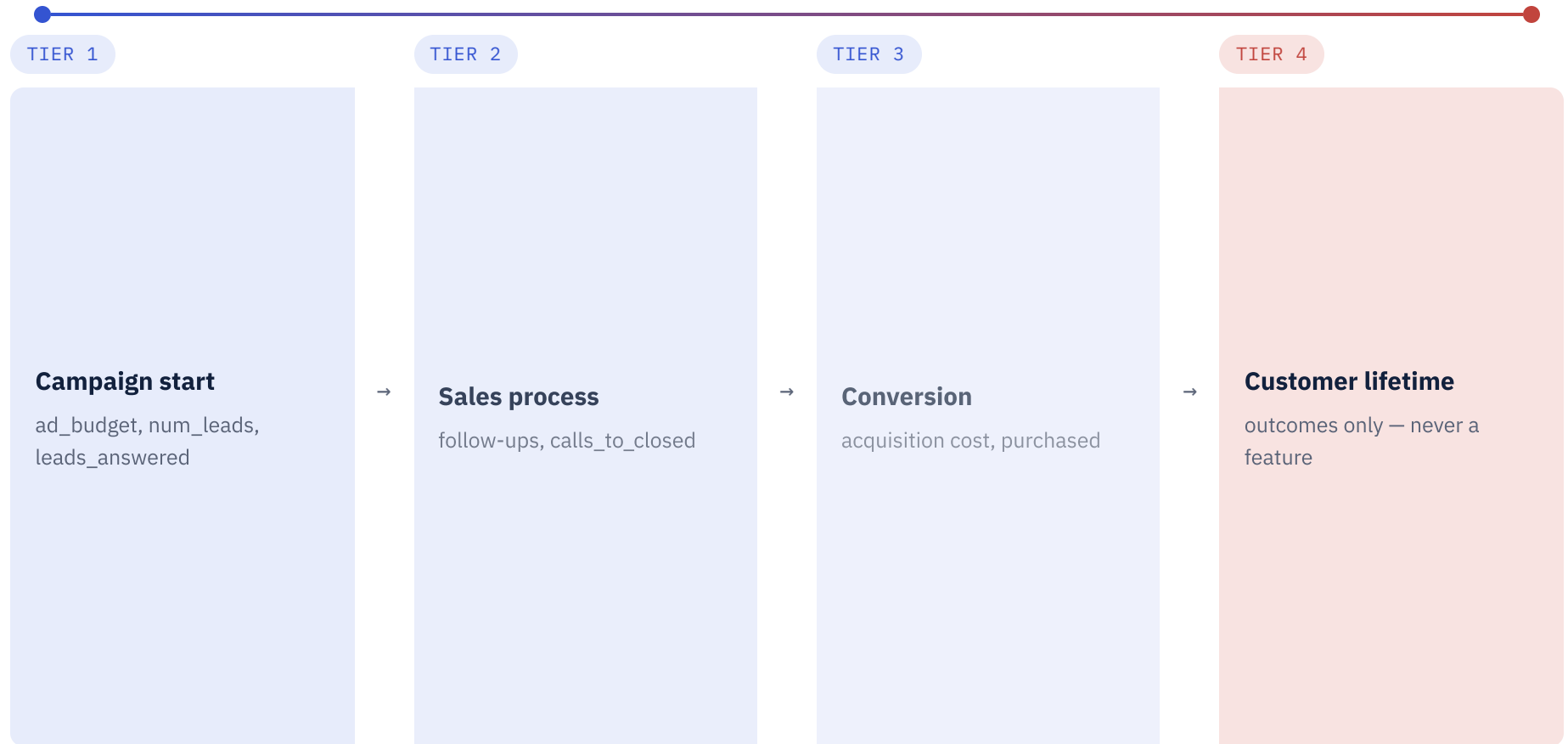
— PROJECT PROCESS

From raw data to a deployed tool



Leakage: a question of timing, not correlation

"Is this value already known at the moment I make the prediction?" — if no, it can't be a feature, no matter how strongly it correlates.



DECISION

cumulative_profit correlates with LTV at $r = 0.85$ — excluded anyway from the LTV model. It's a lifetime outcome, not an input: using it would be a near-circular shortcut.

From features to business answers

LIFETIME

How long will this customer stay?

2.87 mo RMSE

vs. 12.43 mo naive baseline

Key signal: calls_to_closed

Higher calls-to-close is associated with shorter predicted lifetime — a signal for stronger onboarding.

UPSELL

Who should be prioritized for upsell?

0.785 ROC-AUC

recall **83%** model vs. **59%** simple rule

Key signal: calls_to_closed, CAC

A cheap rule captures real signal — but misses too many genuine upsell customers.

SUPER-CUSTOMER

Which campaigns produce high-value customers?

~75% of importance

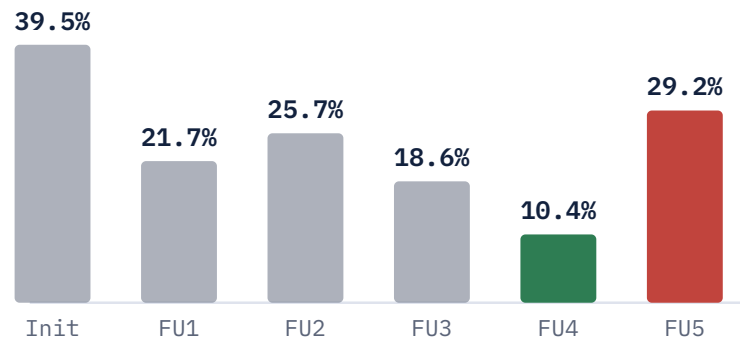
sits in one feature: budget_tier

Mid-budget campaigns score highest

Corrected mid-build: the available features are **campaign-level**, not personal — it scores the campaign a lead came from, not the lead.

"Stop after follow-up 3?"

The data says no.



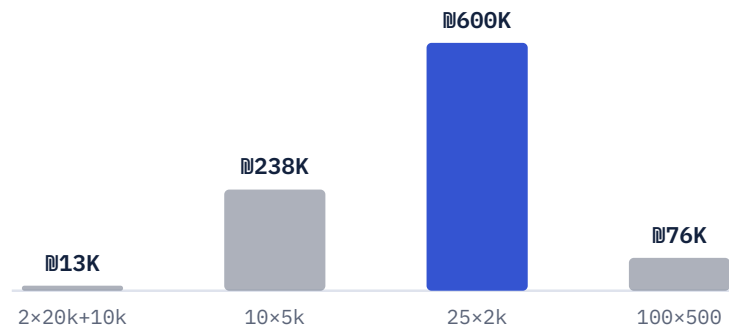
Dropout right after round 3 is the lowest in the sequence — the real leak is entering round 5. Don't shorten the sequence.

The CSV can't explain *why* round 5 leaks — the right next step is an operational check, not more slicing of the same data.

Same ₪50K.

Very different *predicted* outcomes.

MODEL-SIMULATED, NOT REALIZED PROFIT



~46× difference in predicted profit. ~25 × ₪2,000 campaigns is the simulator's top pick.

Neither maximum concentration nor maximum spread wins — the signal is campaign size.

Different models. Different targets. Same signal.

FEWER CALLS TO CLOSE



LIFETIME ↑

UPSELL LIKELIHOOD ↑

PROFIT ↑

MID-BUDGET CAMPAIGNS (≈2K–5K)



CONVERSION ↑

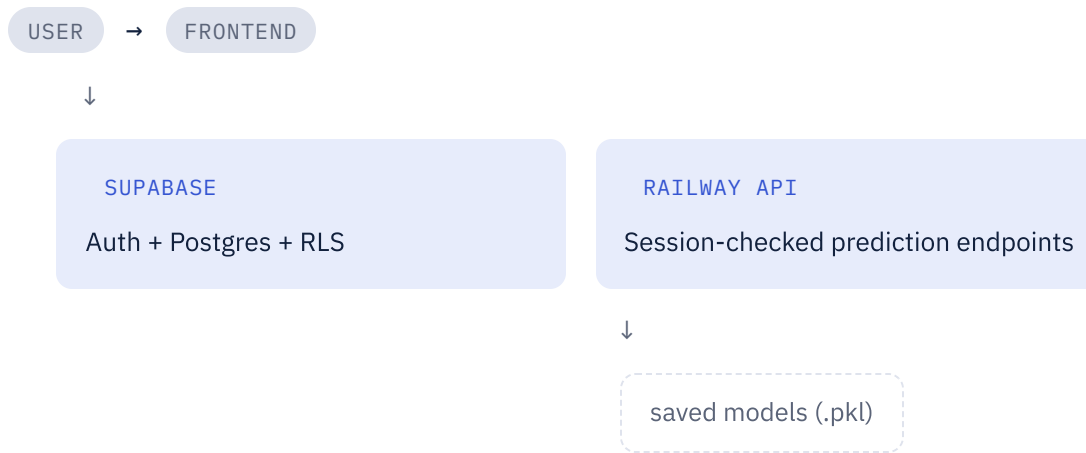
HIGH-VALUE / REFERRAL LIKELIHOOD ↑



*"Across independently trained models and analyses,
the same two patterns repeatedly emerged."*

OBSERVATIONAL PATTERN – NOT EVIDENCE OF CAUSALITY

This didn't stop at a notebook



GITHUB branch → PR → CI → merge

Engineering note: GridSearchCV's clone() failed on CatBoost's categorical features → manual stratified CV.

Authenticated

Every prediction requires a valid session

4 Live Endpoints

Prediction & simulation, deployed

CI / PR Workflow

Every merge shipped through review

- Models trained offline; Railway only serves.
- `service_role` key never touches Railway.

From prediction to action

The next extension: an LLM-generated recommendation layer that turns a model's output and its context into a practical next step.

PLANNED · NOT YET IMPLEMENTED



ILLUSTRATIVE EXAMPLE — NOT A LIVE FEATURE

Model output: Predicted LTV 8 months · unusually high calls-to-close

Recommended action: onboarding + an early retention check-in